

Bijan Pourriahi

Senior Software Engineer | Developer Platforms

API Systems · Developer Tooling · Rust · Python

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PROFILE

Senior software engineer with 10+ years of experience building and operating developer-facing platforms, API integrations, and data-intensive software systems. I own systems end-to-end, from architecture and API design through deployment, observability, and ongoing operation. Recent work includes Rust/Python runtime infrastructure, WebSocket and third-party API integrations, workflow automation, and MCP/CLI tools for AI-assisted development.

I focus on making complex systems easier to understand, debug, and operate. My background spans SaaS platforms, acquired-system integrations, financial infrastructure, automation, and technical leadership.

SELECTED PLATFORM CONTRIBUTIONS

Developer platforms: Built Rust/Python runtime and tooling components powering research, evaluation, and execution workflows with clear interfaces, reproducibility, debuggability, and operational visibility.

API integrations: Designed API, WebSocket, authentication, and adapter layers across external services with different state and failure models.

Workflow systems: Built scheduled jobs, replay pipelines, large-scale evaluations, result capture, and operational review workflows.

AI-assisted development: Integrated Codex CLI, Claude Code, MCP interfaces, and CLI tooling into engineering workflows, focusing on effective tool boundaries, developer experience, and debugging.

PROFESSIONAL EXPERIENCE

Confidential Financial Systems Client — Senior Software Engineer / Technical Lead, Platform & Runtime Systems

2021-Present | Remote

- Owned architecture, implementation, deployment, and operation of a proprietary Rust/Python platform that made complex research and execution workflows reproducible, debuggable, observable, and easier to operate.
- Designed shared Rust/Python research and runtime components so historical simulations and live evaluation workflows could use the same core semantics, including replay, simulation engines, account-state models, risk controls, broker-integration logic, and evidence-capture tooling.
- Designed and implemented API, WebSocket, and integration layers for brokers, prop firms, market-data providers, execution APIs, authentication flows, command/control workflows, dashboards, and operational tooling across external systems with different reliability characteristics.
- Designed MCP interfaces and CLI tools enabling AI agents and developers to interact with internal services, choosing the appropriate abstraction based on remote exposure, local simplicity, standardization, and debuggability.
- Developed workflow systems for scheduled experiments, replay jobs, large-scale evaluations, result capture, and operational review.
- Created experiment pipelines to evaluate thousands of configurations, compare system behavior, and capture reproducible results.
- Owned deployment, monitoring, alerting, health-check, and operational review workflows using Git, Docker, AWS/Linux environments, Grafana, Prometheus, and production runbooks.

Rare Isle Fund — Principal Software Engineer

Apr 2019-Jun 2021 | Remote

- Principal engineer for a small hedge fund, responsible for core automation, analytics, dashboards, data pipelines, and production operations.
- Built automation, analytics, research, and operations infrastructure from scratch, replacing manual workflows with repeatable production software systems.
- Designed Python/Pandas/SQL pipelines for ingesting market data, generating research datasets, running high-volume parameter sweeps and historical simulations, storing results, ranking strategy variants, and reviewing statistical performance.
- Built operational controls, analytics dashboards, monitoring tools, and automated review systems to improve decision speed, production visibility, and system reliability.
- Owned implementation and operations across Python, Pandas, Go, SQL, AWS, Grafana, Linux, DevOps, data pipelines, and production infrastructure.

FocusVision — Senior Software Engineer

Nov 2015-Jun 2018 | Portland, OR

- Senior engineer in a market research SaaS company with multiple acquired production systems and integration needs.
- Led engineering work to integrate five acquired companies/systems into FocusVision, consolidating Rails/PostgreSQL SaaS systems into a more unified production platform.
- Led and mentored engineers while delivering SaaS features, integration work, APIs, database-backed workflows, bug fixes, and production improvements.
- Managed deployments and coordinated with sysadmin/operations teams across production releases, infrastructure changes, data migrations, integration work, and customer-impacting fixes.
- Owned architecture and delivery across requirements analysis, backend/application changes, testing, debugging, documentation, production support, and cross-team coordination.

ADDITIONAL EXPERIENCE

24tru — Software Engineer

Apr 2013-Jun 2015 | Lakewood, CO

- Worked in a small market research SaaS team building software for research collection, data processing, search, and video-based workflows.
- Co-owned the company software platform with one other developer, covering production maintenance, data migrations, new features, customer-impacting fixes, and day-to-day technical operations.
- Managed deployment of the company software stack from the ground up, including release processes, production maintenance, and operational reliability.
- Improved search/Solr indexing performance by roughly 100x by replacing a slow indexing workflow with a custom Java indexing application.
- Built APIs and product features that helped customers collect, upload, process, transcribe, search, and manage video research data.

gloo — Software Developer

Feb 2012-Oct 2012 | Boulder, CO

- Worked in a digital agency/client-services environment.
- Implemented Ruby on Rails features, fixed bugs, wrote tests, and supported PostgreSQL-backed client product development.

TECHNICAL STACK

Languages: Rust, Python, SQL, Ruby, Go, TypeScript, Java

Systems / APIs: API design, WebSockets, REST integrations, command/control flows, CLIs, internal tools, browser tools, GraphQL ecosystem, federation patterns

AI tooling: MCP servers/tools, Codex CLI, Claude Code, agent workflow design, tool orchestration, AI-assisted debugging/review/documentation

Platform: Docker, AWS, Linux, Git, CI/CD, Grafana, Prometheus, Airflow, Dagster, Argo Workflows

Data: PostgreSQL, ClickHouse, SQLite, Pandas, event logs, replayable workflows, query optimization, experiment tracking